



DayOfWeek DayOfMonth Month 2XXX
XX.XX am/pm – XX.XX am/pm
(Duration: 90 minutes)

DEGREES OF MSc, MSci, MEng, BEng, BSc, MA and MA (Social Sciences)

Machine Learning & Artificial Intelligence for Data Scientists

(Answer all of the 3 questions)

This examination paper is worth a total of 60 marks

INSTRUCTIONS TO INVIGILATORS

**Please collect all exam question papers and exam answer scripts and retain for school to collect.
Candidates must not remove exam question papers.**

Question 1: Regression (Total marks: 20)

Consider using regression to predict the world population growth rate using the data shown in the following figure:

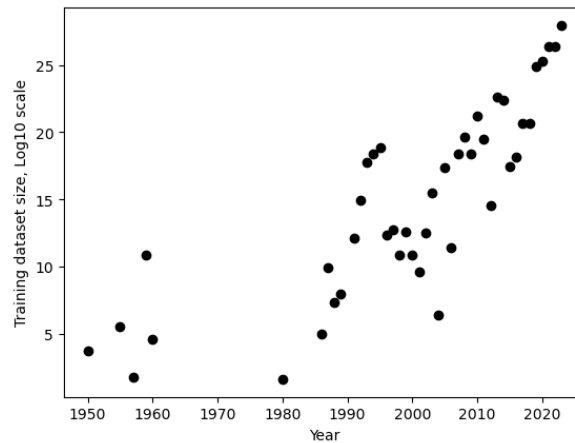


Figure 1.1: Size of training data used in machine learning models from 1950-2023. Source: Modified from <https://ourworldindata.org/grapher/artificial-intelligence-number-training-datapoints>

- (a) A rescaling method was used to rescale the years to values displayed in Figure 1.2. Describe which rescaling method was used (with enough details of the procedure), and why you think it was applied.

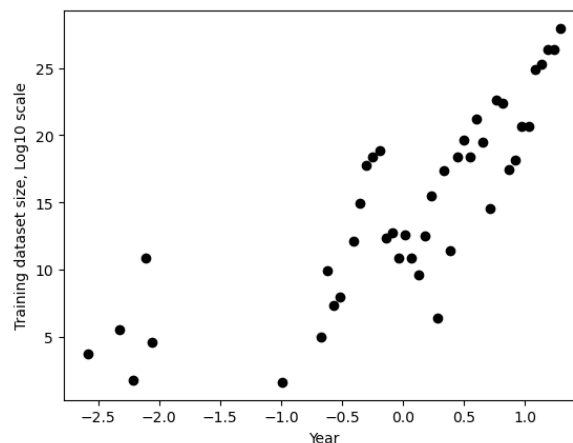


Figure 1.2: Size of training data used in machine learning models with rescaled years.

[4 marks]

Whitening (or equivalent answer) was applied [1]. 1 mark for the correct procedure, $\text{year} - \text{mean}(\text{year}) / \text{std}(\text{year})$. 2 marks for reasoning, the mean of rescaled years in Figure 2 is around 0, indicated by a roughly equal number of data points around 0 [1]. The rescaled years range from around -2.5 to around 1.5, mimicking the unit variance (equals to 1) expected with standard normal distribution [1].

- (b) Consider fitting the data with a polynomial regression of order 2, identify the two most likely poorly fitted data points (use years in Figure 1.1 as reference) and explain why.

[6 marks]

2 marks for identifying the correct poorly fitted data points, several options: (x = 1980, y ≈ 1), (x ≈ 2005, y ≈ 6), 3 points with x range from 1990 to 1995.

4 marks for reasoning: polynomial regression model with an order of 2 is a quadratic (or convex) curve [1]. The global minima of the curve would be in years before 1970 [1]. The right tail end would be following the upward trend from 1980 to 2020. [1]. The curve would closely follow a relatively high-density data region in the middle 1980 to 2020, where points line up almost like a straight line, leaving clear outliers [1].

- (c) Consider fitting the data in Figure 1.1 with a regression with the radial basis function:

$$h_{n,k} = \exp\left(-\frac{(x_n - \mu_k)^2}{2s^2}\right), n = 1, \dots, N; k = 1, \dots, K,$$

Outline one advantage and disadvantage of using this radial basis function over polynomials with the data in Figure 1.1.

[4 marks]

Advantage: the data is not equally distributed across x values, there is a clear gap (1960-1980) [1]. Using location-specific basis functions such as the radial basis functions can model this localized effect by placing basis functions around regions where we have strong data support [1].

Disadvantage: need to choose hyperparameter μ_k and s [1]. A bad choice of μ_k could lead to very bad prediction in the gap (1960-1980) [1].

- (d) Suppose we use the radial basis function in (c), with μ_k set to be x_n and $s = 10$, to fit the data. We used two fitting strategies, namely ridge regression and lasso, and obtained the following fitting models in Figure 1.3 A and B. Identify which fitting strategy is used in each figure and explain why and how the chosen fitting method could have generated the result. (note, each method is used only once).

[6 marks]

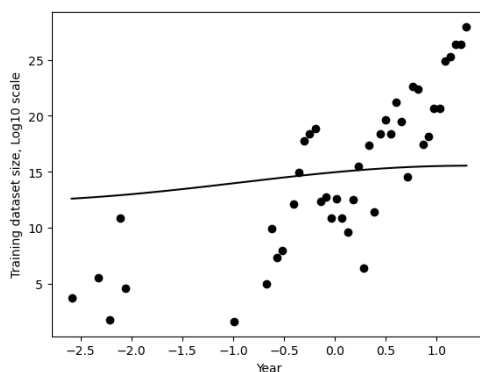


Figure 1.3 A

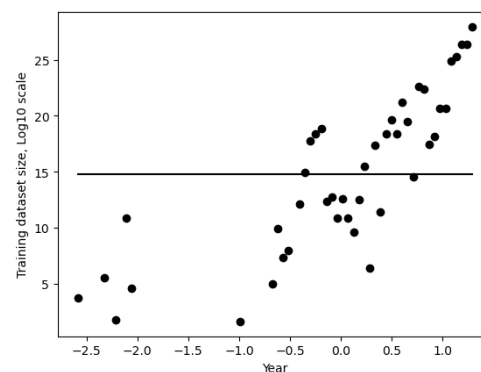


Figure 1.3 B

Figure 1.3 A is fitted with ridge regression [1]. Reasoning: The fitted model is a curve that is close to a flat straight line. This happens when most of the basis functions have close to zero coefficients [1]. This also suggests that the hyperparameter controlling the strength of l2 regularization is set to be too strong [1].

Figure 1.3B is fitted with Lasso [1]. Reasoning: The fitted model is a straight line. This happens when only one basis function has a non-zero coefficient [1]. This also suggests that the hyperparameter controlling the strength of l_1 regularization is set to be too strong [1].

Question 2: Classification (Total marks: 20)

- (a) Assume the following training data in the two-dimensional plane of X_1 and X_2 is available (Figure 2.1). The target variables for the points in the red and blue are +1 and -1. We summarise the data as the following tuples: $\langle (2,0), 1 \rangle$, $\langle (0,2), -1 \rangle$, $\langle (0,2), -1 \rangle$, $\langle (3,0), 1 \rangle$, and $\langle (-1,0), -1 \rangle$, respectively. (Note, you can use LATEX notation, for example X_1 , X_2 , α_1 , α_2 and etc)

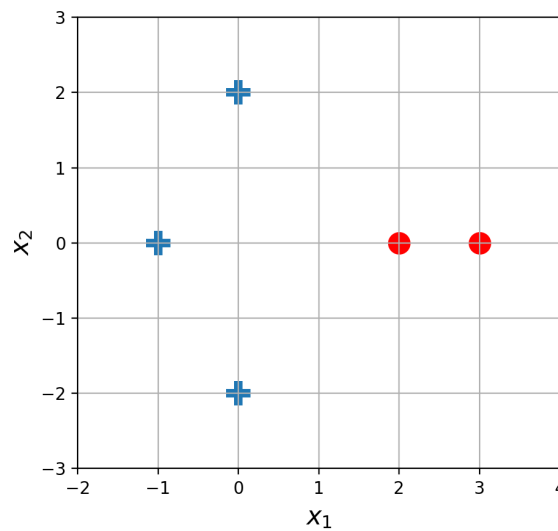


Figure 2.1

- i. Design a k-NN classifier with $k=1$ and write down the equations that specify the decision boundary between the two classes.

[6 marks]

SOLUTION:

$$X_1 - X_2 = 0 \text{ in } X_1 > 0.5 \text{ and } X_2 > 0.5 \quad 2 \text{ marks}$$

$$X_1 + X_2 = 0 \text{ in } X_1 > 0.5 \text{ and } X_2 < -0.5 \quad 2 \text{ marks}$$

$$X_1 - 0.5 = 0 \text{ in } -0.5 < X_2 < 0.5 \quad 2 \text{ marks}$$

- ii. Using the classifier above, determine the class variables C_1 , C_2 , and C_3 for the following test data points: $\langle (0,0), C_1 \rangle$, $\langle (0.7,0), C_2 \rangle$ and $\langle (0.3,0), C_3 \rangle$

[3 marks]

SOLUTION:

$$C_1 = -1, C_2 = 1, C_3 = -1$$

- (b) In the same data set in Figure 2.1, we apply a linear SVM model with the predictor $y(X_1, X_2)$ for classification.

- i. Which data points are the support vectors? Write down the equation for $y(X_1, X_2)$. (Hint: First visually assess the data to determine the decision boundary and the support vectors. Observe the constraints for the margin and SVM classifier.)

[4 marks]

SOLUTION:

(2,0), (0,2), and (0,-2)

1 mark,

$$y(X_1, X_2) = X_1 - 1$$

3 marks

- ii. Specify the Lagrange multipliers $\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5$ for each of the data points in the training data (2,0), (0,2), (0,-2), (3,0), and (-1, 0), respectively. [5 marks]

SOLUTION:

$$\alpha_1 = .5, \alpha_2 = \alpha_3 = 0.25, \alpha_4 = \alpha_5 = 0$$

5 marks

- iii. Which k-NN or SVM classifiers (designed above) will be more accurate? Explain your answer in up to two sentences.

[2 marks]

SOLUTION:

kNN will work better as a nonlinear classifier.

2 marks

3. Unsupervised learning question (Total marks 20)

Consider using the K-means algorithm to perform clustering on the following scenario Figure 3.1 A. We expect to form three clusters as shown in Figure 3.1 B.

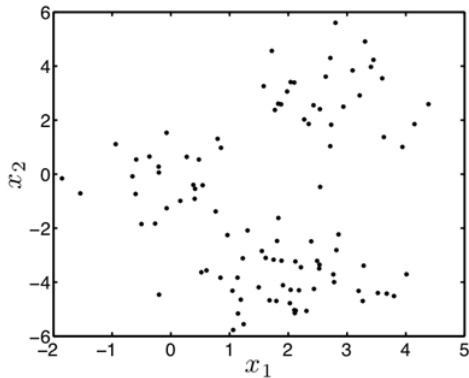


Figure 3.1 A: Original Data

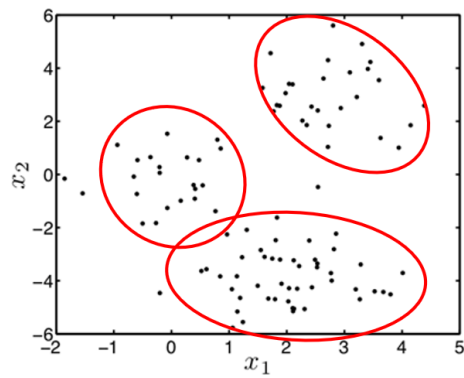


Figure 3.1 B Expected Clusters

- (a) Outline what would happen if we directly apply K-means with Euclidean distance to this data. Can it achieve the clustering objective? How will it split/group the data and why?

[3 marks]

The clusters seem to follow gaussian distributions and some of them are anisotropic (1mark). K-means are not able to encode anisotropy. Therefore, k-means would be partially effective. (1 mark) They will center the three clusters well but they won't be able to capture well the more distance points. (1mark)

- (b) An alternative approach is to use *Kernel K-means*. Would kernel K-means could help in this dataset and why?

[2 marks]

A kernel would help in this case . A gaussian kernel would be able to encode the anisotropy in the data (1mark). In this case, the recovered clusters would have a very close fit with the actual ground truth (1 marks).

- (c) An alternative approach is to use *mixture models*. Would mixture models help to better classify this dataset than K-means and why?

[3 marks]

Mixture models is the ideal method to detect clusters generated via a gaussian model (1 mark). Mixture models assume a gaussian distribution of the underlying clusters and they capture different variations based on the parameters of the distribution (1 mark). In this case, the cluster obtained will be a close fit and you get the additional advantage of recovering the parameters of the generative process (1 mark).

- (d) The plot in Figure 3.2 shows some 2D data. PCA is applied to this data. Explain how the first principal component would look if it is overlaid on the plot. Explain your reasoning. (Hint: if you cannot draw, indicate points/axis on the grid)

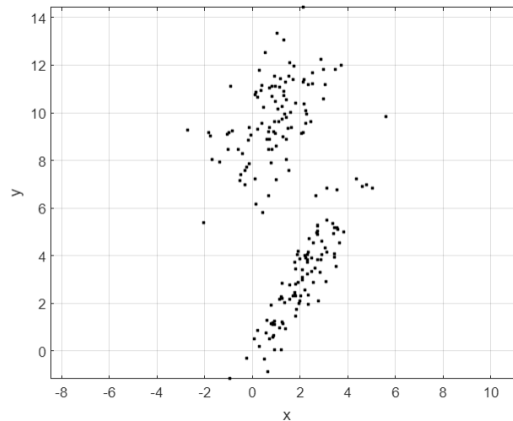
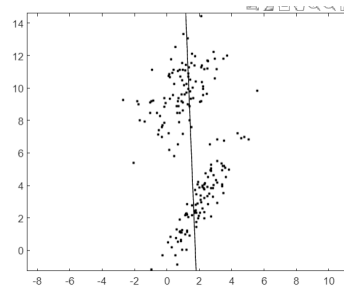


Figure 3.2 2D Points

[2 marks]

First principal component:



The first principal component will be across the direction of the highest variance (1 mark). In this case, we can note that there is a large variation in the y-axis and much smaller than in the x-axis (1 mark).

- (e) Similar to the previous question, explain what the second principal component would look like and why. (Hint: if you cannot draw, you could refer to the x, y-axis for reference)

[2 marks]

PCA decompose the data into orthogonal components (1 mark). Therefore, the second principal component will be orthogonal to the first: (1 mark)

- (f) Explain how you would choose the number of clusters in an application of mixture models.

[3 marks]

Use of k-fold cross-validation. [1 mark]

Estimate the log-likelihood across an increasing number of clusters, K. [1 mark]

Decide the number of clusters based on the extreme/maximum point of the curve. [1 mark]

- (g) Explain how you could detect an outlier point with mixture models. Write a high-level pseudo code and describe each step. (Hint: Start with the expectation-maximization algorithm and how it can facilitate the detection)

[5 marks]

```
--Use gaussian mixture models for clustering:
--Decide on the number of clusters K.
--Use expectation maximization algorithm
---theta={pi,mi,sigma}. These are the parameters of the gaussian distribution
encoding each of the K clusters.
---Initialise parameters, ie. use k-mean to get an approximate answer.
---Expectation step (estimate the probability to assign data to each component qnk)
---Maximisation step (Update mi_k, sigma_k and pi_k)
---Repeat until convergence has been reached
```

[2 marks]

```
--The probability assigned to each data point reflects its distance from the cluster
distribution.
```

[1 mark]

```
--Use a threshold to highlight points that are unlikely to belong to the specific
distribution. (pval= thres)
--Outliers O=points with qnk<thres
```

[2 mark]